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5(a)	Since the standard cell is balanced, the p.d. across XY, $V_{XY} = 1.02 \text{ V}$ (i.e. e.m.f. of the standard cell).	[1]
5(b)	Resistance between XY, $R_{XY} = 50.0 + (10.0 \times (800 / 1000)) = 58.0 \Omega$ Current in the potentiometer wire, $I =$ $= 0.01758 = 0.0176 \text{ A}$	[1] [1]
5(c)	Resistance between AZ, $R_{AZ} = \frac{10 \times 742}{1000} = 7.42 \Omega$ The p.d. across AZ, $V_{AZ} = I R_{AZ} = (0.01758)(7.42) = 0.130 \text{ V}$ Since the unknown cell is balanced, its e.m.f., $E = 0.130 \text{ V}$	[1] [1] [1]
5(d)	p.d. per unit length of potentiometer wire AB = $V_{AZ} / \text{length AZ} = (0.130) / 742$ $= 1.76 \cdot 10^{-4} \text{ V mm}^{-1}$	[1]
5(e)(i)	As the driver circuit current is 17.6 mA, Accumulator e.m.f. = IR_T $2.00 = (0.01758)(R + 50 + 10)$ $R = 53.76 \wedge = 53.8 \wedge$	[2] [1]
5(e)(i)	Based on potential divider rule, when R increases, the potential difference across XB decreases. Thus, the potential difference per unit length for potentiometer wire AB decreases. Hence, the balanced length AY increases to maintain p.d. across AY at 1.02 V (i.e. e.m.f. of standard cell)	[1] [1]
	Max Marks	12

Question	Answer	Marks
6(a)	The magnitude of the induced e.m.f. is directly proportional	[1]